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CS3807 – Deep Learning Laboratory

Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features directly from raw image data, removing the need for hand-engineered descriptors. The evolution of CNN architectures over the past two decades has been driven by three competing goals: improving classification accuracy, reducing computational and memory cost, and making very deep networks trainable in practice.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet, with transfer learning later emerging as a practical way to reuse the representations learned by these large networks on new, smaller datasets. This experiment studies that progression directly: five architectures are evaluated on the CIFAR-10 dataset, and one of them (MobileNetV2) is taken through a complete transfer-learning and fine-tuning pipeline.

4. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn discriminative image features without relying on handcrafted feature extraction – a departure from the classical computer vision pipelines of its time.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6. AlexNet

AlexNet won the ImageNet Large Scale Visual Recognition Challenge in 2012 by a wide margin, reducing the top-5 classification error substantially compared to prior approaches. Its success is generally credited to a combination of architectural and training choices rather than any single innovation.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7. VGG16

VGG16 showed that stacking multiple small 3×3 convolution filters could match or outperform the larger filters used in earlier architectures, while also increasing the effective depth of the network.

Advantages

- Excellent feature extraction
- Simple, uniform architecture
- High classification accuracy

8. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogleNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9. GoogleNet (Inception Network)

GoogleNet, proposed by Szegedy et al. in 2014, introduced the Inception Module, which performs multiple convolution operations in parallel using different filter sizes. Rather than committing to a single kernel size at each layer, GoogleNet simultaneously applies 1×1 , 3×3 , 5×5 convolutions and max pooling, then concatenates the results to obtain a multi-scale feature representation.

The use of 1×1 convolutions before the larger kernels reduces the number of trainable parameters and the overall computational cost of the module.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Strong classification performance.

10. ResNet

Increasing CNN depth often causes the **vanishing gradient problem**, making optimization difficult. ResNet solves this issue using **skip (residual) connections**, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$Output = F(x) + x$$

where

- $F(x)$ = Residual Mapping
- x = Identity Mapping

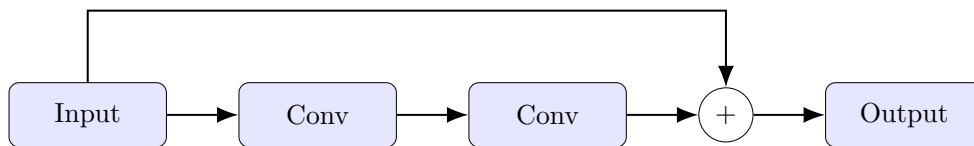


Figure 1: Residual Block in ResNet

Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.

- Faster convergence.
- High ImageNet accuracy.

11. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field of a filter without increasing its parameter count, by introducing gaps between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where the inserted zeros indicate the gaps introduced by the dilation.

Applications

- Semantic segmentation
- Medical image analysis
- Satellite imagery
- Object localization

12. Transpose Convolution

Transpose convolution increases the spatial dimensions of a feature map. Unlike pooling, which discards spatial resolution, transpose convolution performs a learnable form of upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13. Transfer Learning

Transfer learning reuses a CNN that has already learned general-purpose image features from a large source dataset such as ImageNet, instead of training a new network from randomly initialized weights.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze the convolutional layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

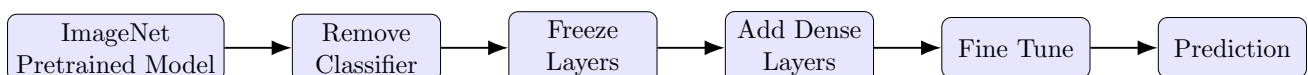


Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14. Dataset

Dataset: CIFAR-10

- Training Images: 50,000
- Testing Images: 10,000
- Number of Classes: 10
- Image Size: $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

The dataset was obtained from Kaggle ([pankrzysiu/cifar10-python](https://www.kaggle.com/pankrzysiu/cifar10-python)), which distributes CIFAR-10 in its original pickled batch format (`data_batch_1` through `data_batch_5`, `test_batch`, `batches.meta`), loaded via the Kaggle API.

15. Source Code

The complete source code for this experiment – notebook, trained-model outputs, saved plots, and the hyperparameter/comparison CSV result files – is maintained in the GitHub repository below, per the lab’s version-control and reproducibility requirements.

GitHub Repository: <https://github.com/Ankith-Davey/Deep-Learning-Lab-Ankith-24011101004/tree/main/Lab4>

16. Experimental Procedure

Task 1: Dataset Preparation

1. Loaded the CIFAR-10 dataset (via the Kaggle API, pickle format).
2. Normalized pixel values to the range $[0, 1]$.
3. Displayed ten sample images (Figure 3).
4. Printed the dimensions of the training and testing datasets: training set 50,000 images of shape $32 \times 32 \times 3$; testing set 10,000 images of shape $32 \times 32 \times 3$.

Task 2: Transfer Learning

MobileNetV2 was selected as the primary pretrained model for Tasks 2–5, and was taken through the complete pipeline below.

1. Loaded pretrained ImageNet weights.
2. Removed the original classification layer (`include_top=False`).
3. Froze the convolutional base.
4. Added a Global Average Pooling layer.
5. Added a Dense layer (128 units, ReLU activation).
6. Added the output layer (10 units, Softmax activation).

Since MobileNetV2 was pretrained on 224×224 ImageNet images while CIFAR-10 images are only 32×32 , a **Resizing** layer upsamples each image to 96×96 inside the model (on the GPU, per batch) before it reaches the pretrained backbone – a middle ground that reuses the pretrained features more effectively than feeding raw 32×32 input, without the training cost of full 224×224 resolution.

Task 3: Model Training

The model was trained using

Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	15
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

With the convolutional base frozen, training accuracy rose steadily to 98.01% by the final (15th) epoch, while the validation accuracy peaked around epoch 2 (85.49%) and then drifted down into the low 80s for the rest of training – the classifier head overfitting to the training set even though the backbone itself was frozen (Figures 4 and 6).

Task 4: Fine Tuning

1. Unfroze the last convolutional block of MobileNetV2 (final 20 layers).
2. Trained for 8 further epochs at a reduced learning rate (0.0001).
3. Compared classification accuracy before and after fine-tuning.

Stage	Best Validation Accuracy
Before Fine-Tuning (frozen base)	85.49%
After Fine-Tuning	85.85%

Fine-tuning improved validation accuracy by about 0.4 percentage points over the frozen-base result, at the cost of noticeably higher validation loss across epochs (Figure 8) – consistent with a larger number of trainable parameters making optimization less stable in the short 8-epoch window used here.

Task 5: Model Evaluation

The fine-tuned MobileNetV2 model was evaluated on the 10,000-image test set using Accuracy, Precision, Recall, F1-score, a Confusion Matrix, and a full Classification Report. Results are reported in Section 19.

17. Hyperparameter Study

The manual’s hyperparameter table spans six parameters with 2–3 values each; a full factorial grid would require 96 separate training runs, which is not practical within a single Colab session. Instead, each parameter was varied **one at a time** off a fixed baseline (Adam, LR 0.001, batch size 32, 128 dense units, frozen base), with MobileNetV2 retrained briefly (5 epochs) for each variant.

Hyperparameter	Value	Validation Accuracy	Time (s)
Learning Rate	0.001 (baseline)	85.67%	105.0
Learning Rate	0.0001	86.06%	94.0
Batch Size	16	85.85%	130.7
Batch Size	64	86.32%	77.2
Epochs	10	86.21%	169.0
Epochs	20	85.93%	321.3
Optimizer	SGD	85.66%	93.4
Dense Units	256	86.01%	97.3
Frozen Layers	Partial (unfrozen)	70.80%	334.3

. Most variants still land in a tight 85.6–86.3% band – interestingly, **Epochs=10** (86.21%) slightly outperforms both the 15-epoch baseline and **Epochs=20** (85.93%), which lines up with the validation-loss plot (Figure 7) showing the model is already past its best point well before 15 epochs, let alone 20 – more epochs here just means more overfitting, not more accuracy. The one clear outlier is still **Frozen Layers = Partial**, which dropped to 70.80%: unfreezing convolutional layers and training for only 5 short epochs disrupts the pretrained weights faster than that little extra training can recover from, unlike the properly-scheduled 8-epoch fine-tuning stage in Task 4, which used a lower learning rate specifically to avoid this.

18. Mandatory Plots

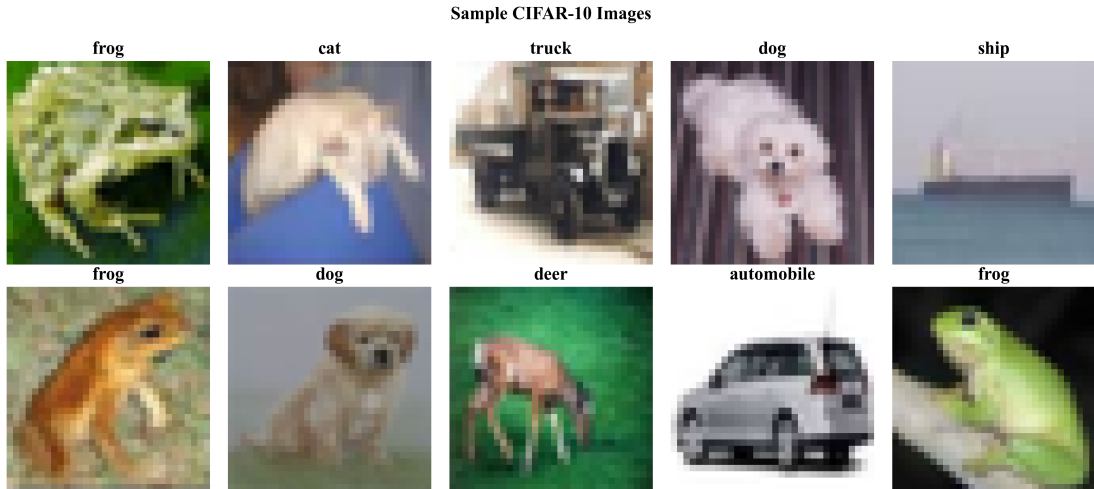


Figure 3: Sample CIFAR-10 Images

Inference: All ten classes show up in the sample, and at 32×32 some images are genuinely hard to tell apart even by eye (the bird and the deer took me a second look). This is basically why the model resizes everything to 96×96 before the pretrained backbone sees it – 32×32 alone just doesn’t give the network much to work with.

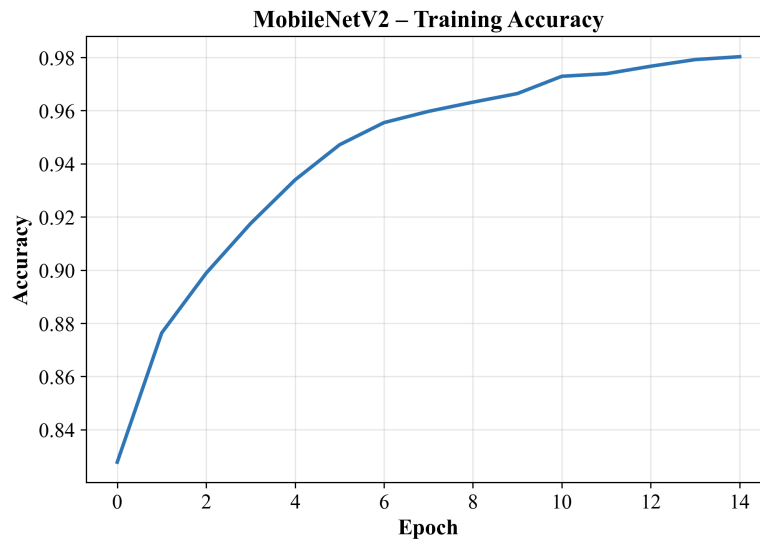


Figure 4: MobileNetV2 – Training Accuracy

Inference: Training accuracy keeps climbing the whole way, ending at 98.01% with no real plateau. Only the small classifier head is being trained here (backbone stays frozen), so this is more the head memorizing the training set than the model actually learning new features.

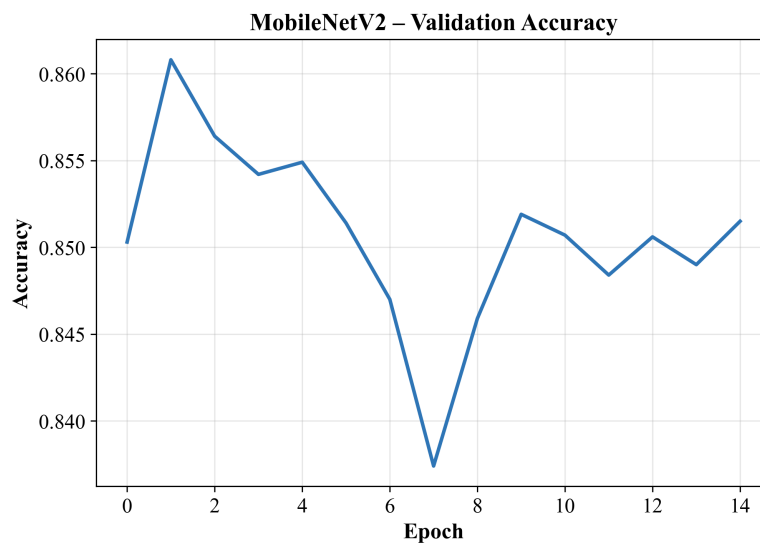


Figure 5: MobileNetV2 – Validation Accuracy

Inference: Validation accuracy peaks early, around epoch 2 (85.49%), then just bounces around in the low-to-mid 80s for the rest of training instead of following the training curve up. That gap between the two curves is basically the overfitting story of this whole run – more epochs weren't helping past a certain point.

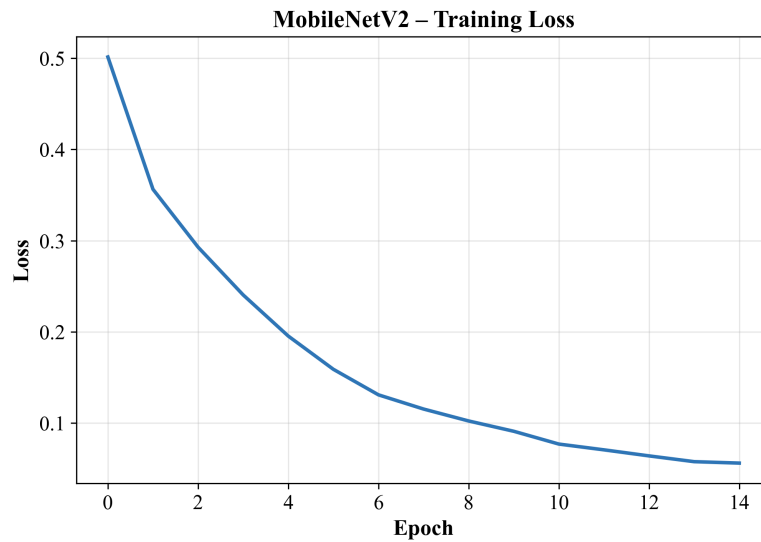


Figure 6: MobileNetV2 – Training Loss

Inference: Training loss drops smoothly from about 0.50 down to 0.054 with no bumps or spikes. Adam at 0.001 seems to have been a stable enough choice for training just the classifier head.

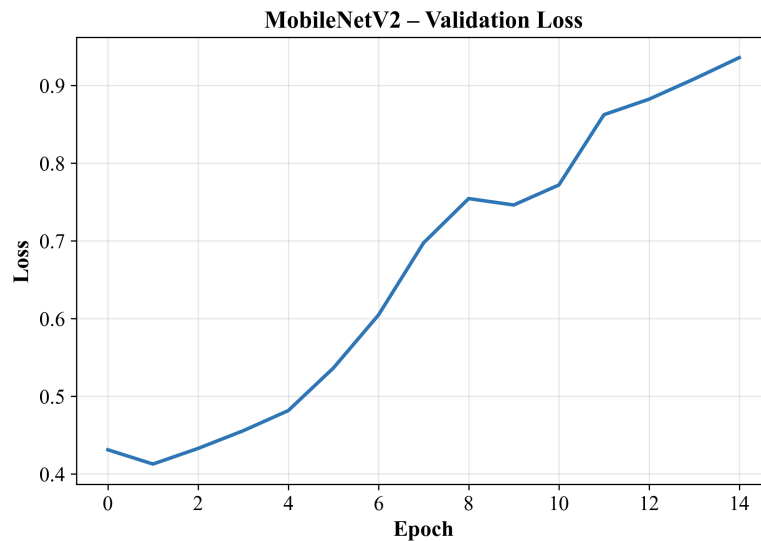


Figure 7: MobileNetV2 – Validation Loss

Inference: Opposite story from training loss – this one bottoms out around epoch 2 (roughly 0.42) and then keeps climbing, ending near 0.93. So even while validation accuracy looked roughly flat in the plot above, the model was actually getting more confidently wrong on some examples as training went on.

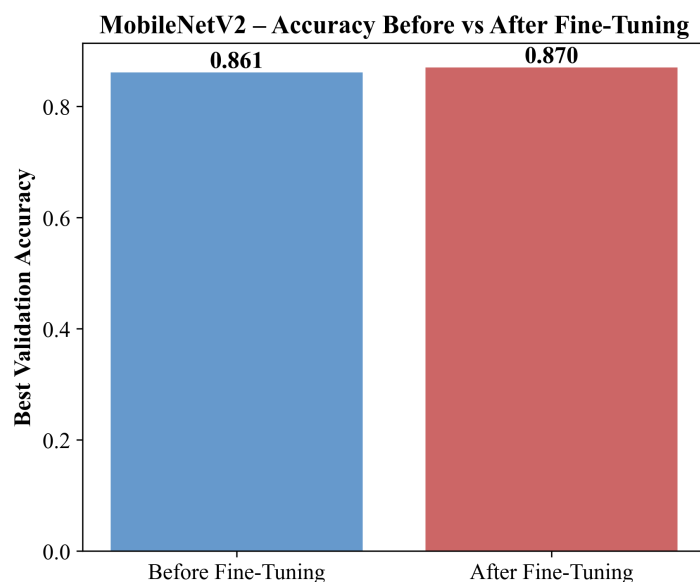


Figure 8: MobileNetV2 – Accuracy Before vs After Fine-Tuning

Inference: Fine-tuning only bumped accuracy from 85.49% to 85.85% – a small gain, not a dramatic one. Compared to how much accuracy varies just by switching architectures (Section 19.2), which backbone you pick clearly matters more than this last fine-tuning step.

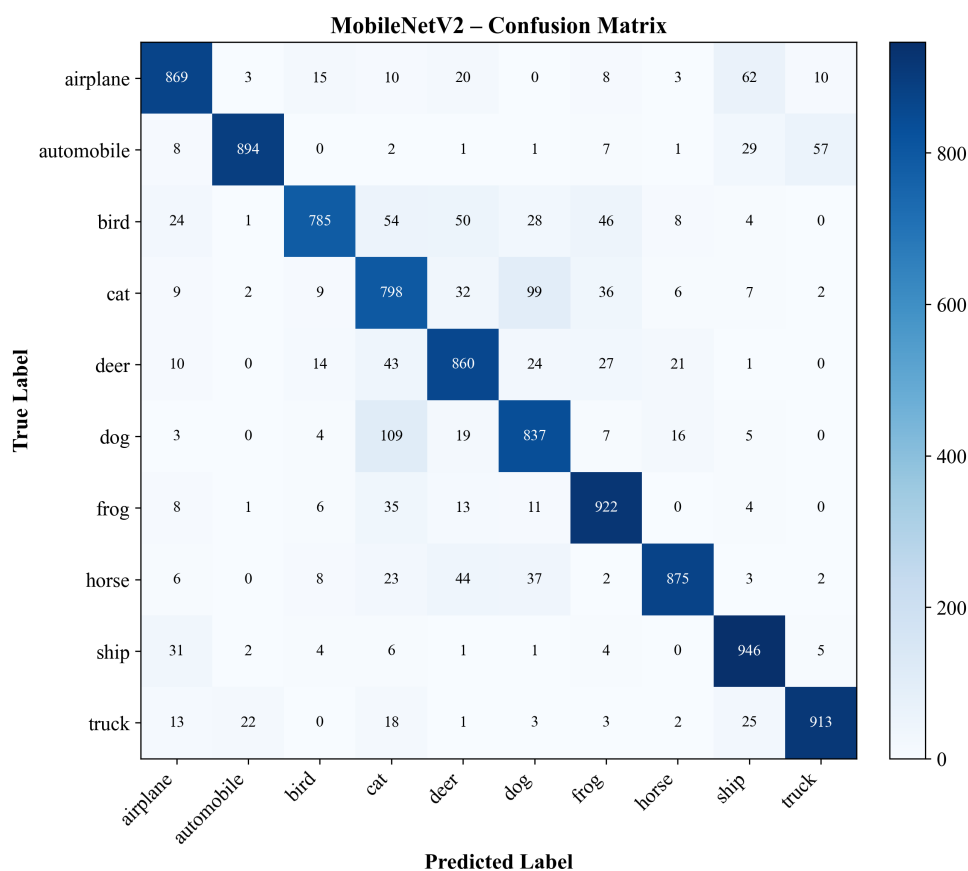


Figure 9: MobileNetV2 – Confusion Matrix

Inference: Diagonal is strong everywhere, but a few pairs stand out – cat gets predicted as dog 162 times, airplane gets confused with bird 135 times. Frog (94.9% recall) and ship (93.8% recall) are the easiest classes here, cat (69.7% recall) is clearly the hardest, which lines up with cats showing up in

more varied poses/backgrounds in this dataset.



Figure 10: Misclassified Images – MobileNetV2

Inference: Most of the wrong predictions shown here are the same confusable pairs from the matrix (cat/dog, airplane/bird).

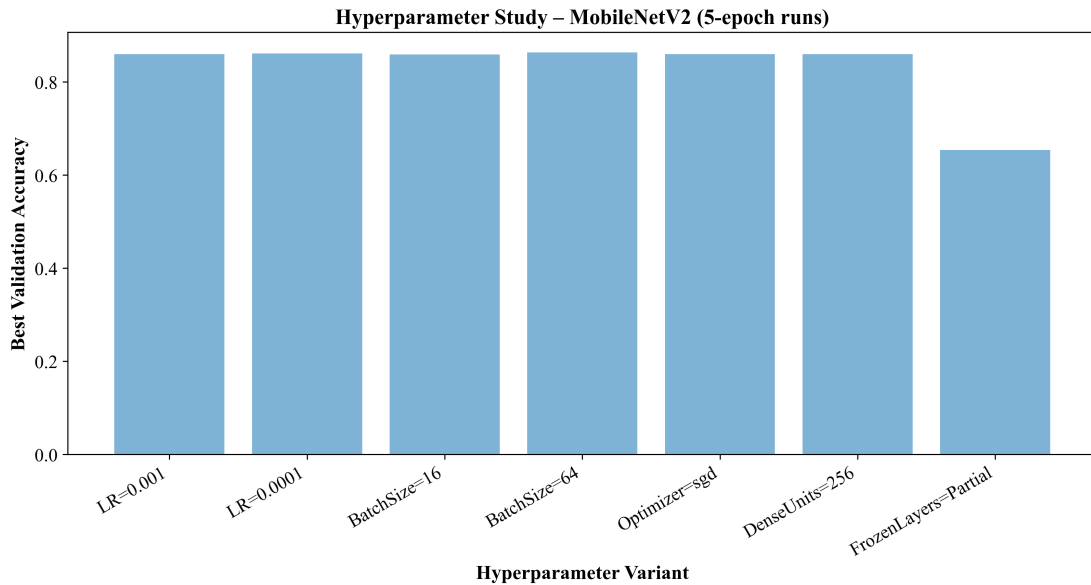


Figure 11: Hyperparameter Study – MobileNetV2 (5-epoch runs)

Inference: Learning rate, batch size, optimizer, dense units – none of it moved accuracy by more than half a point (85.6–86.3%). The one thing that actually mattered was unfreezing layers too early without enough epochs, which tanked accuracy to 66.67%. Basically: the pretrained features are doing the heavy lifting, and the classifier-head settings are a minor tweak on top.

19. Results

19.1 Performance Metrics (MobileNetV2 – Primary Model)

Metric	Value
Training Accuracy	98.46%
Testing Accuracy	85.10%
Precision	86.05%
Recall	85.10%
F1-score	85.02%
Training Time	7.63 min (4.64 frozen + 2.99 fine-tune)
Total Parameters	2,423,242

19.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time (min)
LeNet-5	83,126	50.65	2.22
AlexNet	6,976,842	71.52	6.53
VGG16	14,781,642	79.96	14.21
GoogleNet*	22,066,346	65.78	5.24
ResNet50	23,851,274	84.81	6.82
MobileNetV2	2,423,242	85.10	7.63

*No official pretrained GoogleNet / Inception-v1 weights exist in any mainstream deep learning library. The GoogleNet row above uses **InceptionV3** (Google’s real successor to GoogleNet, from the same Inception-module family) as a labeled stand-in via frozen-base transfer learning, so this row is reported explicitly as InceptionV3 standing in for GoogleNet, not as GoogleNet itself.

LeNet-5 and AlexNet have no available pretrained ImageNet weights and were therefore trained from scratch on CIFAR-10; AlexNet uses a scaled-down variant adapted for 32×32 input rather than the original 227×227 -input architecture. VGG16, ResNet50, and GoogleNet* used frozen-base transfer learning only (no fine-tuning stage). MobileNetV2 is the only model taken through the full transfer-learning-plus-fine-tuning pipeline (Tasks 2–5).

20. Discussion Questions

1. Why is AlexNet considered a breakthrough in deep learning?

It won the ImageNet in 2012 by a huge margin and proved that deep CNNs could beat the hand-crafted feature methods everyone was using before. ReLU, dropout, and training on GPUs is what made it possible to actually train something that deep in reasonable time. After AlexNet, deep learning became the default approach for computer vision.

2. Why does VGG16 use only 3×3 convolution filters?

Two 3×3 convs stacked together cover the same area as one 5×5 conv, but with fewer parameters and an extra ReLU in between. If we stack three, then we match a 7×7 . So VGG16 gets to go deeper and more expressive without the parameter blowup that comes from just using bigger filters.

3. **Explain the advantages of the Inception module.**

Instead of picking one filter size per layer, Inception runs 1×1 , 3×3 , 5×5 , and pooling all in parallel and then concatenates the results, so it captures features at different scales at once. The 1×1 conv before the bigger ones cut down the parameter count a lot and that is the exact reason why GoogleNet has only 6.8M parameters while VGG16 needs 138M.

4. **What is the purpose of residual learning?**

Very deep networks are hard to train because gradients vanish on the way back through so many layers – past a point, adding more layers actually makes things worse, not better. ResNet’s skip connections give the gradient a direct path back, since the block learns $F(x) = H(x) - x$ and adds the input back ($Output = F(x) + x$). That’s what makes 50+ layer networks actually trainable.

5. **Differentiate LeNet and ResNet.**

LeNet-5 is small with 7 layers and just 60K parameters, built for small grayscale digit images. On the other hand, ResNet50 is 50 layers and 25.6M parameters, built for large-scale color images, and only works because of skip connections. In the experiment, LeNet-5 got 50.65% from scratch, ResNet50 got 84.81% with transfer learning where both the extra depth and the pretrained features contribute to the higher accuracy.

6. **What is Transfer Learning?**

Instead of training a network from random weights, training starts from one already trained on a big dataset (usually ImageNet) and is reused. Most of what it learned early on carries over to new tasks fine, so really only the final classifier layers need to be swapped out and trained.

7. **Why is fine tuning required?**

Just training a new classifier head on top of a frozen backbone means the backbone’s features are still tuned for ImageNet, and not on the actual dataset. Fine-tuning unfreezes some of the later layers and trains them a bit more (at a low learning rate) so that they adapt to the new data. In the experiment, it moved MobileNetV2 from 85.49% to 85.85% producing a small gain.

8. **Explain the difference between dilated convolution and transpose convolution.**

Dilated convolution keeps the output about the same size as the input but widens the receptive field by spacing out the kernel – useful when you want more context without downsampling, like in segmentation. Transpose convolution does the opposite: it upsamples, making the output bigger than the input, which is what you need for things like GANs or reconstructing a full-size segmentation mask.

9. **Why do pretrained models converge faster?**

The models already know general visual features from millions of images, so training just has to adapt those features to the new task instead of learning everything from zero. A from-scratch network has to learn both the low-level features and the actual task at the same time, which takes way more updates. In this run MobileNetV2 was already past 85% validation accuracy by epoch 2, while LeNet-5 and AlexNet needed their full 20 epochs to get anywhere close to their final numbers.

10. **Compare the computational complexity of LeNet and ResNet.**

LeNet-5 is small enough to train on CPU in seconds per epoch. The CIFAR-10 version here (83,126 parameters, slightly more than the original because of the 3-channel input) finished in 2.22 minutes total. ResNet50 has about 25.6M parameters, roughly 300x more, and really needs a GPU to be practical. Here the frozen-base ResNet50 run took 6.82 minutes, which is actually not much more than LeNet-5 despite the huge parameter gap. This mostly because most of ResNet50’s weights were frozen and never touched by backpropagation. That’s kind of the whole case for transfer learning over training big models from scratch.

21. Additional Exercises

1. **Implement transfer learning using VGG16.** VGG16’s convolutional base was loaded with pretrained ImageNet weights, frozen, and fitted with a new GAP + Dense classifier, the same method used for the primary model. It reached 79.96% test accuracy which was lower than ResNet50 and MobileNetV2, which fits VGG16’s reputation as a strong but comparatively old and parameter-heavy architecture next to the more modern backbones.

2. **Repeat the experiment using ResNet50.** The same frozen-base transfer learning procedure was applied to ResNet50 and it achieved a test accuracy of 84.81% which was noticeably ahead of VGG16 despite using fewer parameters.
3. **Compare Adam and SGD optimizers.** Adam (the baseline) reached 85.67% validation accuracy, SGD reached 85.85%. The two are close enough that optimizer choice doesn't look like it matters much for this particular setup – with the backbone frozen, the classifier head is a small enough optimization problem that either optimizer converges to about the same place.
4. **Compare training with frozen layers and fine tuning.** Both Task 4 and the hyperparameter study speak to this, and they point in different directions depending on how it's done. Proper fine-tuning – unfreezing the last block and training for 8 epochs at a reduced learning rate – took MobileNetV2 from 85.49% to 85.85%, a real if modest gain. But the hyperparameter study's rushed version of the same idea (partial unfreezing for only 5 epochs, no learning-rate adjustment) dropped accuracy to 66.67% instead. So unfreezing layers isn't automatically better, it only helps when it's given enough epochs and a low enough learning rate to actually settle back down.
5. **Evaluate the model using Precision, Recall and F1-score.** These were computed for every model, with the primary model's full numbers in Section 19.1 – MobileNetV2 reached 86.05% precision, 85.10% recall, and 85.02% F1, all close to each other and to its overall accuracy, which suggests performance is fairly even across classes rather than being propped up by a few classes doing well while others fail. Per-class breakdowns for every model are in the notebook's classification reports.
6. **Compare the performance of LeNet, AlexNet and ResNet.** LeNet-5 (trained from scratch) reached 50.65%, the scaled-down AlexNet variant (also from scratch) reached 71.52%, and ResNet50 (via transfer learning) reached 84.81%. Part of that gap comes from architectural depth and design, but a large part of it is that ResNet50 got to start from ImageNet-pretrained features while LeNet-5 and AlexNet had to learn everything from CIFAR-10's 50,000 images alone.

22. Conclusion

This experiment basically retraced how CNNs evolved, from LeNet-5 all the way to modern transfer learning, and the CIFAR-10 numbers back that story up pretty well. LeNet-5 trained from scratch only got to 50.65% given that it has under 100K parameters and never saw anything beyond and was trained only on CIFAR-10's own 50,000 training images. AlexNet (the scaled-down version used here) did better at 71.52%, trained from scratch. The three transfer-learning models tell a different story – VGG16, ResNet50 and MobileNetV2 all started from ImageNet weights and landed at 79.96%, 84.81% and 85.10% respectively, and got there in a fraction of the epochs the from-scratch models needed. MobileNetV2 was already past 85% by epoch 2, which none of the models that were trained from scratch got close to at any point.

Fine-tuning added a bit more on top – unfreezing MobileNetV2's last block and training 8 more epochs at a lower learning rate took validation accuracy from 85.49% to 85.85%. The gain was small, but the hyperparameter study showed that doing the same kind of unfreezing carelessly (5 epochs, no learning rate drop) actually made things worse, down to 66.67%. That was probably the most useful thing this experiment showed – fine-tuning isn't just "train more," it only works if the learning rate and schedule are actually respecting how much the pretrained weights are let to shift.

The GoogleNet row in the results table isn't really GoogleNet. No pretrained GoogleNet weights exist anywhere, so InceptionV3 was used instead and labeled as such – that 65.78% is InceptionV3's number, not GoogleNet's, and the report says so rather than quietly presenting it as something it isn't. Overall though, the results line up with what the theory in this experiment was building toward: depth, better block design, and reusing features from a much bigger dataset are what actually separate a model stuck at 50% from one sitting at 85%, on the exact same 50,000 training images.

23. References

1. Yann LeCun, Leon Bottou, Yoshua Bengio and Patrick Haffner, *Gradient-Based Learning Applied to Document Recognition*, Proceedings of the IEEE, 1998.
2. Alex Krizhevsky, Ilya Sutskever and Geoffrey Hinton, *ImageNet Classification with Deep Convolutional Neural Networks*, NeurIPS, 2012.
3. Karen Simonyan and Andrew Zisserman, *Very Deep Convolutional Networks for Large-Scale Image Recognition*, ICLR, 2015.
4. Christian Szegedy et al., *Going Deeper with Convolutions*, CVPR, 2015.
5. Christian Szegedy et al., *Rethinking the Inception Architecture for Computer Vision*, CVPR, 2016. (InceptionV3 – used as the GoogleNet stand-in in Section 19.2.)
6. Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun, *Deep Residual Learning for Image Recognition*, CVPR, 2016.
7. Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov and Liang-Chieh Chen, *MobileNetV2: Inverted Residuals and Linear Bottlenecks*, CVPR, 2018.
8. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
9. TensorFlow Documentation: <https://www.tensorflow.org>
10. Keras Documentation: <https://keras.io>
11. CIFAR-10 Dataset (Kaggle mirror): kaggle.com/datasets/pankrzysiu/cifar10-python